

Data Structure	Size	Construction time	Iteration time
Array	300 Million	3.486 seconds	0.077 seconds
	400 Million	4.555 seconds	0.083 seconds
	500 Million	6.202 seconds	0.123 seconds
	600 Million	7.314 seconds	0.138 seconds
ArrayList	300 Million	9.072 seconds	0.335 seconds
	400 Million	11.909 seconds	0.454 seconds
	500 Million	16.206 seconds	0.616 seconds
	600 Million	20.608 seconds	0.827 seconds
Vector	300 Million	15.339 seconds	7.418 seconds
	400 Million	19.801 seconds	9.657 seconds
	500 Million	27.511 seconds	12.112 Seconds
	600 Million	29.652 seconds	14.551 seconds

Arrays are by far the fastest, and vectors are the slowest. However, with arrays, you need a set size, and you cannot go bigger than the set maximum, you would need to copy all the content over to a new, larger array.

If you know how many elements you will be dealing with, arrays are the best choice, however, if you do not know how many elements there would be, or the number of elements could grow infinitely, then an arraylist would be the best option.